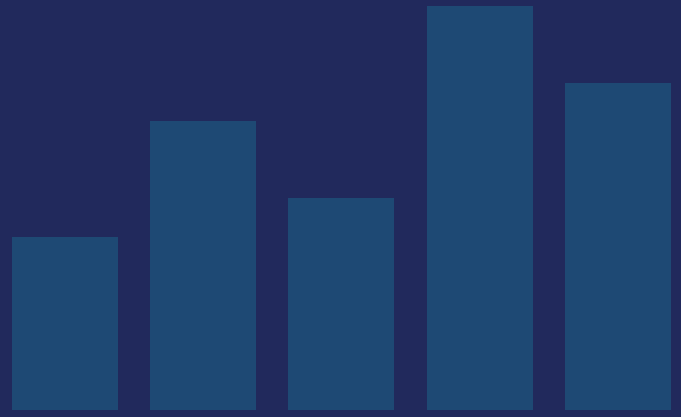
PROJECT CASE STUDY

Telecom Customer Churn Analysis

End-to-End Analytics: SQL → Data Modeling → DAX → Executive Dashboard



Sudulaguntla Yamini | Data Analyst

THE CHALLENGE

Why Churn Matters Here

A telecom provider was losing customers steadily, but had no structured way to answer:

- Which contract types churn the fastest?
- Which services and payment methods correlate with churn?
- How much revenue is actually walking out the door?

This project builds a SQL-to-DAX analytics pipeline that turns raw customer records into an executive dashboard answering exactly that.

26.54%

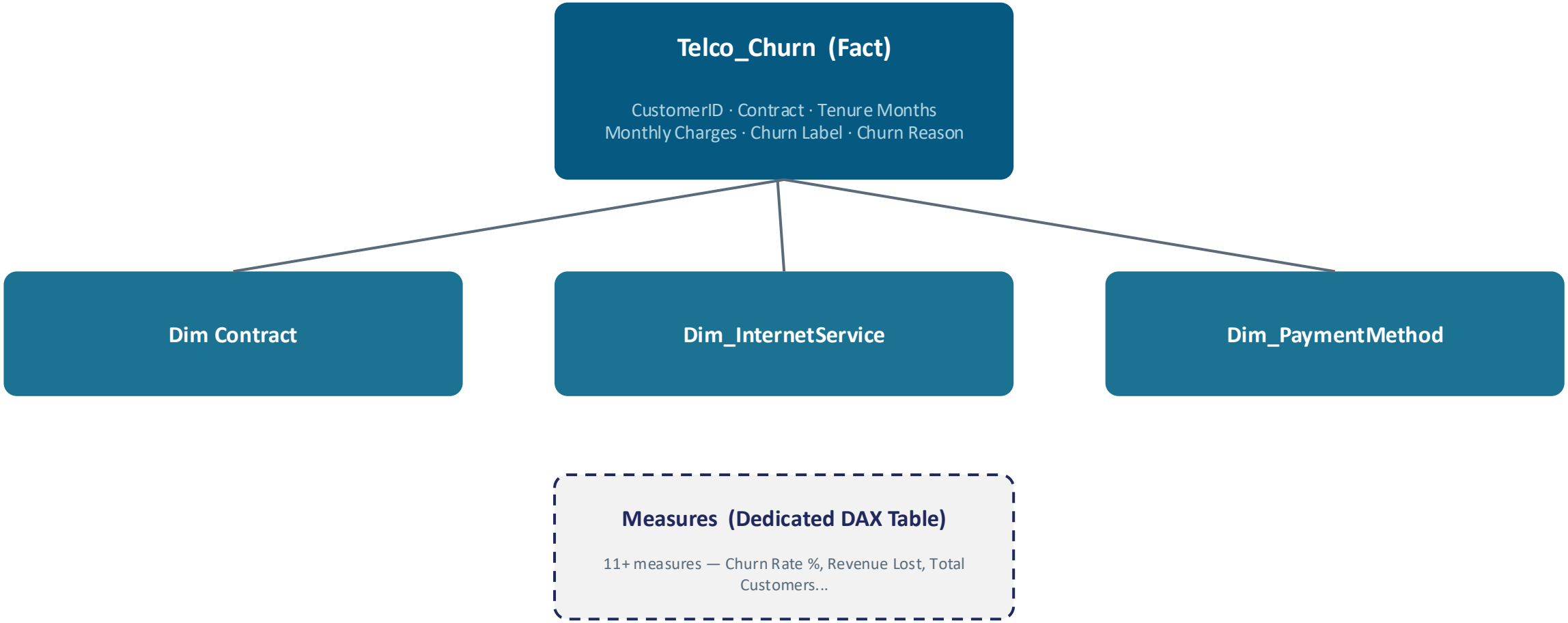
Overall Customer Churn Rate

₹139.13K

Revenue Lost to Churned Customers

DATA MODELING

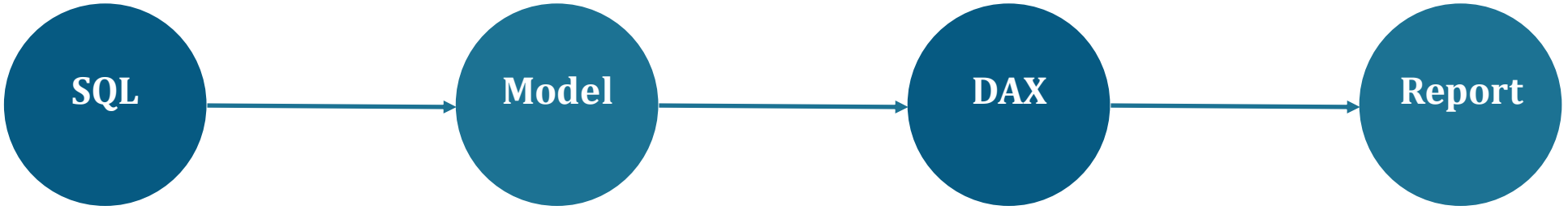
Star-Schema Data Architecture



No relationships — referenced directly by every visual via DAX

WORKFLOW

From Raw Data to Executive Dashboard



Extract & Validate

MySQL queries clean and aggregate raw customer records

Star Schema

Fact + dimension tables related for efficient filtering

Measures

11+ measures power every KPI and chart on the dashboard

6-Page Dashboard

Interactive Power BI report with bookmarks and tooltips

SQL — DATA LAYER

Querying the Raw Data

Churn Rate Calculation

```
SELECT ROUND(
  SUM(CASE WHEN `Churn Label`='Yes'
      THEN 1 ELSE 0 END)*100
  /COUNT(*),2) AS Churn_Rate
FROM telco_customer_churn;
```

Powers the 26.54% churn rate KPI card

Churn Rate by Contract Type

```
SELECT Contract,
  ROUND(SUM(CASE WHEN
    `Churn Label`='yes'
    THEN 1 ELSE 0 END)*100
  /COUNT(*),2) AS ChurnRate
FROM telco_customer_churn
GROUP BY Contract;
```

Revealed month-to-month plans churn far more than annual contracts

Revenue Lost to Churn

```
SELECT ROUND(
  SUM(`Monthly Charges`),2
) AS Revenue_From_Churned
FROM telco_customer_churn
WHERE `Churn Label`='Yes';
```

Quantifies the direct revenue impact — feeds the ₹139.13K KPI

DAX — SEMANTIC LAYER

Key Measures Behind the Dashboard

Total Customers

```
COUNTROWS(Telco_Churn)
```

Churn Customers

```
CALCULATE(COUNTROWS(Telco_Churn), Telco_Churn[Churn Label]="Yes")
```

Churn Rate %

```
DIVIDE([Churn Customers], [Total Customers], 0)
```

Revenue Lost by Churn Customers

```
CALCULATE(SUM(Telco_Churn[Monthly Charges]), Telco_Churn[Churn
Label]="Yes")
```

Active Customers

```
[Total Customers] - [Churn Customers]
```

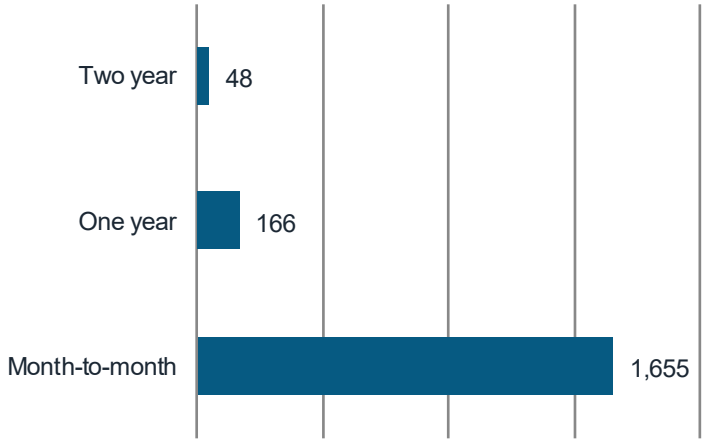
Representative formulas derived from measure names & report bindings extracted from the .pbix — swap in your exact DAX from Power BI's Model view for full accuracy.

THE DASHBOARD

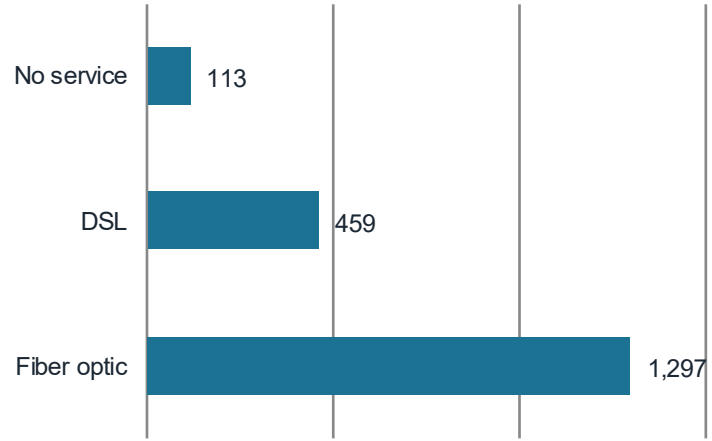
Executive Overview — Headline Numbers

7K	5K	2K	26.54%	₹139.13K
Total Customers	Active Customers	Churn Customers	Churn Rate	Revenue Lost to Churn

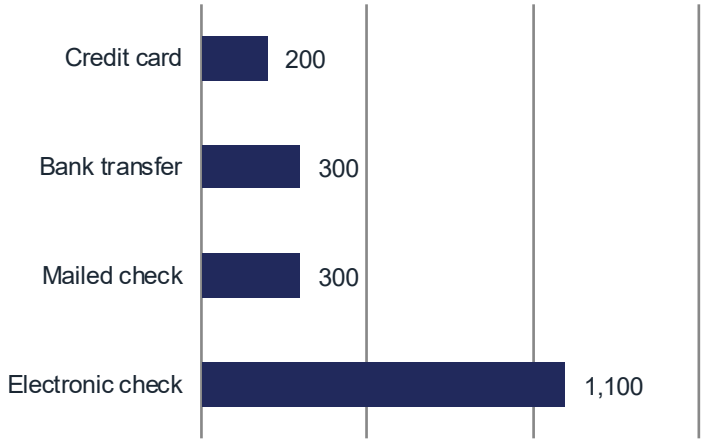
Churned by Contract



Churned by Internet Service

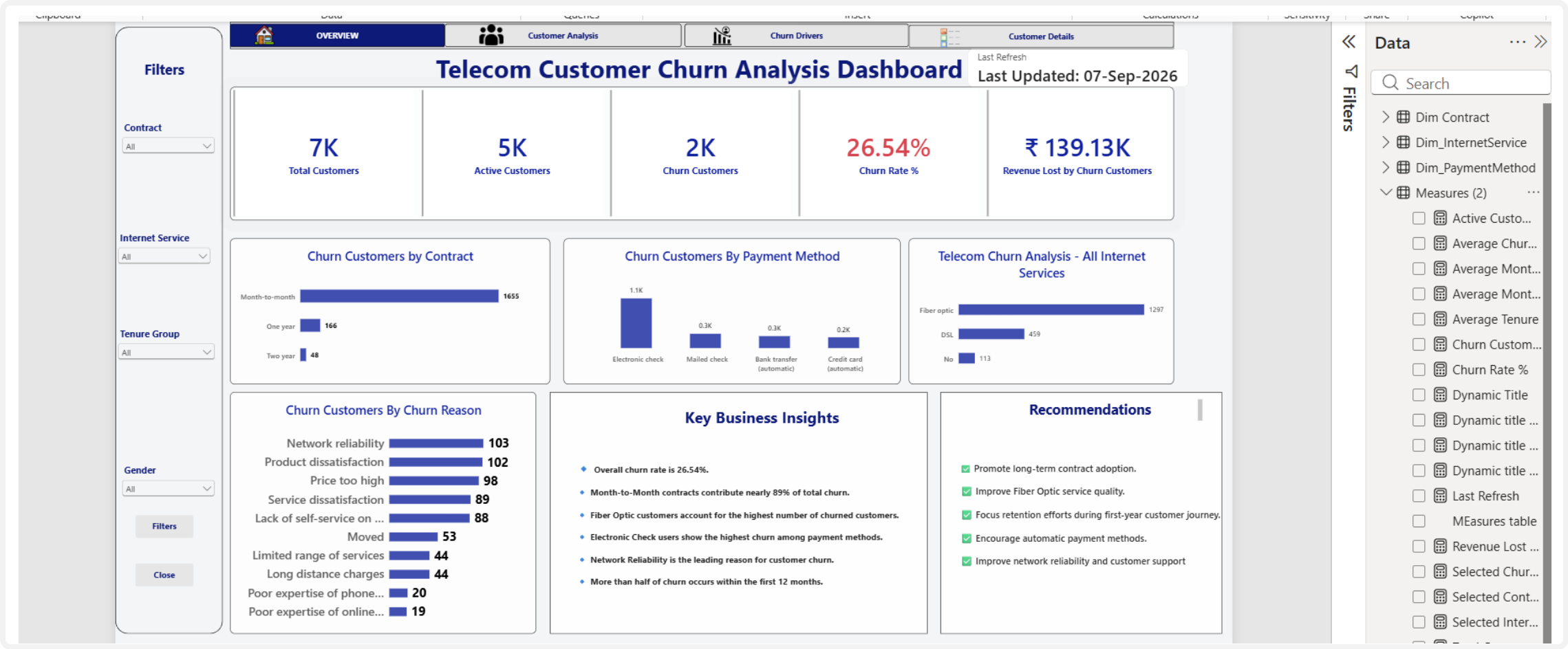


Churned by Payment Method



THE DASHBOARD

Live From Power BI — Executive Overview



FINDINGS

Key Business Insights

1. Overall churn rate is 26.54% across the customer base
2. Month-to-month contracts contribute nearly 89% of total churn
3. Fiber Optic customers account for the highest number of churned customers
4. Electronic Check users show the highest churn among payment methods
5. Network reliability is the leading reason customers give for leaving
6. More than half of all churn happens within a customer's first 12 months

BUSINESS IMPACT

Recommendations

- ✓ Promote long-term contract adoption to reduce month-to-month churn exposure
- ✓ Improve Fiber Optic service quality — the largest churn segment by service type
- ✓ Focus retention efforts during the first-year customer journey
- ✓ Encourage automatic payment methods over manual electronic checks
- ✓ Improve network reliability and customer support responsiveness

SKILLS DEMONSTRATED

- SQL
- Power BI
- DAX
- Data Storytelling

Let's Connect

yaminisudulaguntla@gmail.com
linkedin.com/in/yamini-sudulaguntla
github.com/Yamini-Sudulaguntla

Open to Data Analyst / Power BI Developer opportunities.

